

List of Contents

Declaration	i
Acknowledgement	ii
Abstract	iii
List of Figures	vii
List of Tables	ix
List of Abbreviations	x
1 Introduction	1
1.1 Overview	1
1.2 Background and Motivation	1
1.3 Research Questions / Problem Statements	3
1.4 Research Objectives	3
1.5 Scope of the Research	4
1.6 Key Contributions	5
1.7 Organization of the Report	6
2 Literature Review	8
2.1 Introduction	8
2.2 Existing Research	8
2.2.1 Stereo Matching Fundamentals	8
2.2.2 The Deep Learning Transition	13
2.2.3 The 3D Cost-Volume Era	14
2.2.4 Iterative Refinement	15
2.2.5 Foundation-Model Era	17
2.2.6 Requirements of Edge-Deployable Stereo Matching	19
2.2.7 Classification of Efficient Stereo Methods	19
2.2.8 Backbone Substitution	21
2.2.9 Cost-Volume Compression	21
2.2.10 Iterative-Loop Compression	21
2.2.11 Knowledge Distillation	22
2.2.12 Architectural Compression and Adaptive Compute	23
2.3 Research Gap	24
2.4 Comparative Analysis	25
2.4.1 Cross-Domain Generalization	25
2.4.2 Benchmarks and Datasets	26
2.4.3 Accuracy, Parameters, and Latency Across Method Families	27
2.5 Summary	28

3	Engineering Design and Methodology	30
3.1	Introduction	30
3.2	Research Methodology	30
3.3	Design Requirements and Requisites	31
3.4	Design Criteria	32
3.5	Design Steps	32
3.6	Modeling of Proposed System	32
3.7	Modeling of Conceptual Design	34
3.8	Preliminary Design	35
3.9	Detailed Design	36
3.9.1	Shared Feature Encoder and Context Encoder	36
3.9.2	Tile Initialization and Cost Volume	38
3.9.3	Recurrent Tile Refinement	38
3.9.4	Plane-Aware Cross-Scale Propagation	40
3.9.5	Geometry Encoding Volume and Fail-Soft Fusion	41
3.9.6	Plane Rendering and Learned Convex Upsampling	42
3.9.7	From Disparity to 3D Reconstruction	43
3.10	Design Assessment and Performance Indices	44
3.11	Summary	45
4	System Development and Implementation	46
4.1	Introduction	46
4.2	Software Development	46
4.3	Network Implementation	47
4.4	Optimization and Configuration Selection	47
4.5	Training Objective and Protocol	49
4.5.1	Training Objective	49
4.5.2	Datasets and Training Protocol	50
4.6	Hardware and Embedded Deployment	52
4.7	Principle of Operation	52
4.8	Summary	53
5	Experimental Setup and Data Collection	55
5.1	Introduction	55
5.2	Experimental Setup and Test Bench	55
5.3	Calibration and Rectification	56
5.4	Datasets	58
5.4.1	Synthetic Training and Test Corpus: SceneFlow	58
5.4.2	Zero-Shot Cross-Domain Benchmark: Middlebury 2014	58
5.4.3	Physical Stereo Rig Captures	59
5.5	Data Collection	59
5.6	Image Preprocessing Pipeline	59
5.7	Experimental Procedure and Evaluation Metrics	61
5.8	Summary	62

6	Results and Discussion	64
6.1	Introduction	64
6.2	Experimental Results	64
6.2.1	Training Dynamics	64
6.2.2	In-Domain Accuracy: SceneFlow Test	65
6.3	Comparison	66
6.3.1	Comparison with Published Methods on SceneFlow	66
6.3.2	Zero-Shot Cross-Domain: Middlebury 2014	67
6.4	Error Analysis	68
6.4.1	Robustness to Rectification Imperfection	68
6.4.2	Qualitative Results	69
6.4.3	3D Reconstruction of Real Scenes	70
6.4.4	Comparison with LiDAR Point Clouds	71
6.5	Validation	72
6.5.1	Design-Space Exploration of the Tile Chassis	72
6.5.2	Architecture Ablation of the Recurrent Family	73
6.5.3	Training-Recipe and Efficiency Studies	74
6.6	Performance Evaluation	76
6.6.1	Efficiency and Deployment Measurements	76
6.7	Discussion	76
6.8	Summary	77
7	Societal, Environmental Impact and Sustainability	78
7.1	Introduction	78
7.2	Impact of the Thesis on Societal, Health, and Cultural Issues	78
7.3	Safety and Ethical Considerations	79
7.4	Alignment with Sustainable Development Goals (SDGs)	80
7.5	Summary	81
8	Thesis Management and Finance	82
8.1	Introduction	82
8.2	Management of the Thesis	82
8.2.1	Time Frame / Gantt Chart	83
8.3	Overall Budget	84
8.3.1	Component and Software Budget	84
8.4	Resource Planning	85
8.5	Individual and Team Work Contribution	86
8.6	Summary	86
9	Conclusions, Limitations and Future Works	88
9.1	Conclusion	88
9.2	Limitations	88
9.3	Future Works	89
	References	90